

ISRO 2025: Topic-Wise Practice Questions

TOPIC 2: OPERATING SYSTEMS

Based on 2023 ISRO Questions: 9, 48, 49, 50, 51, 52, 53, 54

****Coverage Analysis****: This topic had 8 questions (10%) in 2023 paper, making it the second highest weightage area.

****Key Concepts Tested****: Memory allocation strategies, process scheduling, synchronization mechanisms, multithreading models, interrupts, system calls.

MEMORY MANAGEMENT (Questions 1-15)

****1. Which heap memory allocation strategy exploits spatial locality best:****

- (a) Best Fit
- (b) First Fit
- (c) Next Fit
- (d) Worst Fit

****2. In paging, logical address is divided into:****

- (a) Page number and page offset
- (b) Segment number and offset
- (c) Base and limit
- (d) Frame number and offset

****3. Page size of 4KB means offset requires how many bits:****

- (a) 10
- (b) 12
- (c) 14
- (d) 16

****4. Internal fragmentation occurs in:****

- (a) Paging
- (b) Segmentation
- (c) Dynamic partitioning
- (d) Linked allocation

****5. Which memory allocation suffers from external fragmentation:****

- (a) Fixed partitioning
- (b) Paging
- (c) Dynamic partitioning
- (d) Both (a) and (c)

****6. Translation Lookaside Buffer (TLB) is used for:****

- (a) Page table caching
- (b) Process scheduling

- (c) File system caching
- (d) Network buffering

****7. Effective access time formula with TLB hit ratio h , TLB time t , memory time m :****

- (a) $h \times t + (1-h) \times (t+m+m)$
- (b) $h \times (t+m) + (1-h) \times (t+m)$
- (c) $h \times t + (1-h) \times m$
- (d) $h \times m + (1-h) \times (t+m)$

****8. Virtual memory allows:****

- (a) Program larger than physical memory
- (b) Multiple programs in memory
- (c) Faster program execution
- (d) All of above

****9. Demand paging loads pages:****

- (a) All at program start
- (b) Only when referenced
- (c) In sequential order
- (d) Based on priority

****10. Thrashing occurs when:****

- (a) Too many page faults
- (b) System runs multiple processes
- (c) Memory is insufficient
- (d) CPU utilization is high

****11. Working set model helps in:****

- (a) Page replacement
- (b) Determining memory requirements
- (c) Process scheduling
- (d) File allocation

****12. FIFO page replacement can suffer from:****

- (a) Optimal behavior
- (b) Belady's anomaly
- (c) Stack property
- (d) LRU behavior

****13. Which page replacement algorithm is optimal:****

- (a) FIFO
- (b) LRU
- (c) Optimal
- (d) Clock

****14. LRU page replacement requires:****

- (a) Reference bit
- (b) Modify bit

- (c) Usage timestamp
- (d) Valid bit

****15. In segmentation, logical address consists of:****

- (a) Page number, offset
- (b) Segment number, offset
- (c) Frame number, offset
- (d) Block number, offset

**PROCESS MANAGEMENT (Questions 16-30)**

****16. Process states in order of typical lifecycle:****

- (a) New → Ready → Running → Waiting → Terminated
- (b) Ready → New → Running → Waiting → Terminated
- (c) New → Running → Ready → Waiting → Terminated
- (d) New → Waiting → Ready → Running → Terminated

****17. Process Control Block (PCB) contains:****

- (a) Process ID, program counter, registers
- (b) Memory management information
- (c) I/O status information
- (d) All of above

****18. Context switching involves:****

- (a) Saving current process state
- (b) Loading new process state
- (c) Updating process tables
- (d) All of above

****19. fork() system call:****

- (a) Creates child process
- (b) Replaces current process
- (c) Waits for child termination
- (d) Terminates current process

****20. exec() system call:****

- (a) Creates child process
- (b) Replaces current process image
- (c) Waits for child termination
- (d) Terminates current process

****21. wait() system call is used by:****

- (a) Child to wait for parent
- (b) Parent to wait for child
- (c) Process to wait for I/O
- (d) Scheduler to wait for CPU

****22. Zombie process is:****

- (a) Process waiting for I/O
- (b) Process in ready state
- (c) Terminated process with entry in process table
- (d) Process with infinite loop

****23. Orphan process is:****

- (a) Process without parent
- (b) Process without child
- (c) Process waiting for I/O
- (d) Process in suspended state

****24. Inter-Process Communication methods:****

- (a) Shared memory
- (b) Message passing
- (c) Pipes
- (d) All of above

****25. Thread vs Process:****

- (a) Threads share address space
- (b) Threads have separate address space
- (c) Processes share address space
- (d) No difference

****26. User-level threads are managed by:****

- (a) Operating system kernel
- (b) Thread library
- (c) Hardware
- (d) Scheduler

****27. Kernel-level threads are managed by:****

- (a) Thread library
- (b) Operating system kernel
- (c) User process
- (d) Application

****28. Many-to-one threading model:****

- (a) Many user threads to one kernel thread
- (b) One user thread to many kernel threads
- (c) One-to-one mapping
- (d) Dynamic mapping

****29. One-to-one threading model:****

- (a) Many user threads to one kernel thread
- (b) Each user thread maps to kernel thread
- (c) One-to-many mapping
- (d) No mapping

****30. Which threading model does Linux use:****

- (a) Many-to-one
- (b) One-to-many
- (c) One-to-one
- (d) Many-to-many

PROCESS SCHEDULING (Questions 31-45)

****31. Long-term scheduler:****

- (a) Selects processes from ready queue
- (b) Controls degree of multiprogramming
- (c) Performs context switching
- (d) Manages memory allocation

****32. Short-term scheduler:****

- (a) Controls degree of multiprogramming
- (b) Selects process from ready queue for execution
- (c) Handles swapping
- (d) Manages I/O operations

****33. Medium-term scheduler:****

- (a) Controls multiprogramming degree
- (b) Handles swapping operations
- (c) Selects ready processes
- (d) Manages file operations

****34. FCFS scheduling:****

- (a) Preemptive
- (b) Non-preemptive
- (c) Priority-based
- (d) Time-slice based

****35. SJF scheduling gives:****

- (a) Minimum average waiting time
- (b) Maximum CPU utilization
- (c) Best response time
- (d) Fair scheduling

****36. Round Robin uses:****

- (a) Priority
- (b) Time quantum
- (c) Process size
- (d) I/O requirements

****37. Priority scheduling can suffer from:****

- (a) Starvation
- (b) High overhead
- (c) Poor response time
- (d) Low throughput

****38. Aging technique prevents:****

- (a) Deadlock
- (b) Starvation
- (c) Thrashing
- (d) Race condition

****39. Multilevel queue scheduling:****

- (a) Processes can move between queues
- (b) Fixed assignment to queues
- (c) Single queue for all processes
- (d) Dynamic queue creation

****40. Multilevel feedback queue:****

- (a) Fixed queue assignment
- (b) Processes can move between queues
- (c) Single priority level
- (d) No time quantum

****41. Preemptive scheduling:****

- (a) CPU allocated until process terminates
- (b) CPU can be taken away from process
- (c) No context switching
- (d) Processes run to completion

****42. Turnaround time is:****

- (a) CPU burst time
- (b) Waiting time in ready queue
- (c) Completion time - arrival time
- (d) Response time

****43. Response time is:****

- (a) First CPU allocation - arrival time
- (b) Completion time - arrival time
- (c) Total execution time
- (d) Waiting time

****44. CPU utilization is:****

- (a) $\text{Time CPU is busy} / \text{Total time}$
- (b) Number of processes completed
- (c) Average turnaround time
- (d) Average waiting time

****45. Throughput is:****

- (a) CPU utilization percentage
- (b) Number of processes completed per unit time
- (c) Average response time
- (d) Total execution time

SYNCHRONIZATION (Questions 46-55)

****46. Critical section problem requirements:****

- (a) Mutual exclusion, progress, bounded waiting
- (b) Mutual exclusion only
- (c) Progress only
- (d) Bounded waiting only

****47. Peterson's solution works for:****

- (a) Two processes
- (b) Multiple processes
- (c) Any number of processes
- (d) Single process

****48. Semaphore operations:****

- (a) wait() and signal()
- (b) P() and V()
- (c) down() and up()
- (d) All of above

****49. Binary semaphore can have values:****

- (a) 0 and 1
- (b) Any positive integer
- (c) Negative values
- (d) Floating point values

****50. Counting semaphore can have:****

- (a) Only 0 and 1
- (b) Any non-negative integer
- (c) Only positive integers
- (d) Any integer

****51. Producer-Consumer problem uses:****

- (a) One semaphore
- (b) Two semaphores
- (c) Three semaphores
- (d) No semaphores

****52. Reader-Writer problem ensures:****

- (a) Multiple readers can read simultaneously
- (b) Writer needs exclusive access
- (c) Readers and writers are mutually exclusive
- (d) All of above

****53. Dining Philosophers problem demonstrates:****

- (a) Deadlock
- (b) Starvation

- (c) Resource allocation issues
- (d) All of above

****54. Monitor provides:****

- (a) High-level synchronization
- (b) Condition variables
- (c) Mutual exclusion
- (d) All of above

****55. Priority inheritance protocol prevents:****

- (a) Deadlock
- (b) Priority inversion
- (c) Starvation
- (d) Race conditions

**ANSWER KEY - OPERATING SYSTEMS**

****1-10:**** (c), (a), (b), (a), (d), (a), (a), (d), (b), (a)

****11-20:**** (b), (b), (c), (c), (b), (a), (d), (d), (a), (b)

****21-30:**** (b), (c), (a), (d), (a), (b), (b), (a), (b), (c)

****31-40:**** (b), (b), (b), (b), (a), (b), (a), (b), (b), (b)

****41-50:**** (b), (c), (a), (a), (b), (a), (a), (d), (a), (b)

****51-55:**** (c), (d), (d), (d), (b)

**Critical Focus Areas for September 14th:**

1. ****Memory Management (Q1-15):**** Understand paging, segmentation, virtual memory concepts
2. ****Process Scheduling (Q31-45):**** Master different scheduling algorithms and their properties
3. ****Synchronization (Q46-55):**** Critical for ISRO - semaphores, monitors, classical problems
4. ****Process Management (Q16-30):**** Process states, system calls, threading models

****Study Priority:**** Based on 2023 ISRO pattern, focus 30% time on synchronization problems as they appeared frequently.